Prompt for generating summaries

```
In previous interactions, {previous_summary}. The current time and date are {current_date_and_time}.
{speaker_1_name} and {speaker_2_name} talked today and had the following conversation: {session} Summarize the
interactions between {speaker_1_name} and {speaker_2_name} so far. Include key details about both speakers and
include time references wherever possible.
```

Session (input)

1:14 pm on 25 May, 2023

Hey Nate! Haven't talked in a few days. Crazy things happened to me!

Hi Joanna! Long time no see! What's been going on? You sound excited!

Woo! I finally finished my first full screenplay and printed it last Friday. I've been working on for a while, such a relief to have it all done!

Wow, that sounds awesome! What's it about? Glad it's all down! Thanks, Nate! It's a mix of drama and romance!

Thanks, Nate! It's a mix of drama and romance!

Wow, that's amazing! How do you feel now that it's finished? Do you have any new plans for it?

I'm feeling a rollercoaster of emotions - relief, excitement, some anxiety - over finishing this project.

...

Previous summary (input)

On 21 January 2022, Joanna and Nate reunited after a long time without seeing each other. Nate won his first video game tournament playing Counter-Strike: Global Offensive. Joanna enjoys writing, reading, watching movies, and exploring nature as hobbies. They both share a love for movies, with Nate favoring action and sci-fi while Joanna prefers dramas and romcoms. Joanna recommended a romantic drama to Nate, which he expressed interest in watching. Nate praised Joanna's recommendation and promised to give it a try.

New summary (output)

Joanna and Nate reunited on 21 January 2022 after a long time without seeing each other. They share a love for movies, with Nate favoring action and sci-fi while Joanna prefers dramas and romcoms. Joanna recommended a romantic drama to Nate, which he expressed interest in watching. On 23 January 2022, at 2:01 pm, Joanna shared her excitement about finishing her first full screenplay, which is a mix of drama and romance, and her plans to submit it to film festivals. Nate congratulated her and shared his own experience with his pet turtles, recommending having pets for times of stress. Joanna mentioned her allergies to most animals with fur, prompting Nate to inquire further about her allergies and express empathy towards her situation. Despite her allergies, Joanna finds joy in writing and spending time with friends. The conversation ended on a positive note with Nate encouraging Joanna to keep pursuing her passion for writing.

Figure 8: **Prompt for generating conversation summaries.** The prompt used to iteratively generate a summary for the current session by conditioning on summary from preceding sessions and the raw conversation logs of the current session (top); and an example of inputs for the prompt and corresponding output summary of a session from the LoCoMo dataset.

Conversation Statistics	# Counts
Total. # conversations h.	50
Avg. # sessions k. in conversation h	19.3
Avg. # turns j. in session k	15.8
Avg. # tokens. conversation h	9,209.2
Avg. # tokens. dialogue h_{k_j} of turn j in session k	30.2
Avg. # tokens. observation o_{k_j} of turn j in session k	18.2
Avg. # tokens. summary w_k of session k	127.4
QA Benchmark Statistics	
# questions. single-hop retrieval	2,705 (36%)
# questions. multi-hop retrieval	1,104 (14.6%)
# questions. temporal reasoning	1,547 (20.6%)
# questions. open domain knowledge	285 (3.9%)
# questions. adversarial	1,871 (24.9%)
Total. # questions.	**7,512**
Event Summarization Statistics	
Avg. # ground truth events. in conversation h	24.2
Avg. # tokens. event summary	896.5
Multi-modal Dialogue Generation Statistics	
Avg. # images. in conversation h	32.3

Table 5: **Dataset Statistics** of conversation and corresponding benchmark

resolution can be performed effectively by replacing images in LoCoMo with their captions generated using BLIP-2 (Li et al., 2023b), and using state-of-art LLMs to reason over natural language text interleaved with image captions. Hence, our experiments for the question answering and event summarization tasks are conducted using LLMs. We use the images directly only for experiments on the multimodal dialog generation task.

Question Answering. We carry out experiments using three distinct methodologies: (1) **Base** involves utilizing LLMs to directly conduct the task within a constrained context. The task description comes after the dialogue history. To accommodate the restricted context window size, earlier dialogues are omitted; (2) **Long-context** employs LLMs with an extended context window to expose the models to as much dialogue context as possible; (3) **Retrieval-augmented Generation (RAG)** involves retrieving relevant context from a database of dialog history, observations, or session-level summaries. *Observations* are assertions about each speaker extracted from the dialog history as described in §3.3, see an example in Figure 9. Session-level *summaries* are concise summaries of the conversation that takes place in each session, see an example in Figure 8.

For the retrieval model, we employ DRAGON (Lin et al., 2023). In the *Base*, we utilize Mistral-7B (Jiang et al., 2023), LLama-70B-chat (Touvron et al., 2023), `gpt-3.5-turbo` [9], and `gpt-4-turbo` [10]. To assess the effectiveness in practical scenarios for *Long-context* and *RAG*, we draw comparisons using variants of

[9] https://platform.openai.com/docs/models/gpt-3-5

[10] https://platform.openai.com/docs/models/gpt-4-and-gpt-4-turbo

Prompt for generating observations

```
Write a concise and short list of all possible OBSERVATIONS about each speaker that can be gathered from the
CONVERSATION. Each observation should contain a piece of information about the speaker. The OBSERVATIONS should
be objective factual information about the speaker that can be used as a database about them. Avoid abstract
observations about the dynamics between the two speakers such as 'speaker is supportive', 'speaker appreciates'
etc. Do not leave out any information from the CONVERSATION.
```

Conversation (input)

1:56 pm, May 8, 2023

Hey Joanna! Long time no see! What's up? ..

Hey Nate! Long time no see! I've been working on a project lately - it's been pretty cool. What about you - any fun projects or hobbies?

I won my first video game tournament last week - so exciting!

Wow Nate! Congrats on winning! Tell me more - what game was it?

Thanks! it's a team shooter game.

Wow, great job! What was is called?

The game was called Counter-Strike: Global Offensive, and me and my team had a blast to the very end!

Sounds like a fun experience .. if I'm not into games.

...

...

Observations (output)

Joanna

- Joanna has been working on a project recently.
- Joanna enjoys writing, reading, watching movies, and exploring nature as hobbies.
- Joanna is into dramas and romcoms when it comes to movies.
- Joanna recommends a romantic drama movie that is all about memory and relationships.
- Joanna watched the recommended movie around 3 years ago and even owns a physical copy.

Nate

- Nate won his first video game tournament last week.
- The video game Nate won the tournament in is called Counter-Strike: Global Offensive.
- Playing video games and watching movies are Nate's main hobbies.
- Nate enjoys action and sci-fi movies.
- Nate loves watching classics.

Figure 9: **Prompts for generating observations from conversations.** The prompt used to generate observations from a conversation (top); and an example of inputs for the prompt and corresponding output observations for a session from the LoCoMo dataset.

`gpt-3.5-turbo`. We do not report the performance of long-context fine-tuned open-source models (Chen et al., 2023) or those utilizing sliding window (Bertsch et al., 2024; Dao et al., 2022) due to the variability inherent across different open-source models and the potential reduction in their capability on shorter context.

Event Summarization. We present experiments conducted in two distinct configurations. We use both the **Base** and **Long-context** setups from the question answering task, but we refrained from including RAG since summarization requires a comprehensive understanding of the entire dialogue, rather than just retrieving a specific portion. A notable distinction in our approach, compared to the question-answering task, lies in our handling of the context. Specifically, we employ an iterative process of creating a summary of a preceding session and then use that summary as a basis to generate the summary for the subsequent session (Chang et al., 2023). Further, we use a single in-context demonstration of input and output to guide the model toward selecting only significant life events for the summary.

Multi-modal Dialogue Generation. For evaluating multi-modal dialogue generation, we train MiniGPT-5 (Zheng et al., 2023) on 50 conversations generated using our automated pipeline (without human filtering) as detailed in §3. Three distinct versions of the model were developed, each with varying training data: (1) **Base** trains on preceding dialogue turns; (2) **+ summary** trains on both prior dialogue turns and a global summary of the ongoing conversation; (3) **+ observation** trains on both preceding dialogue turns and relevant observations retrieved from the conversation history. For each of these models, we started with a MiniGPT-5 checkpoint pretrained on the MMDialog dataset (Feng et al., 2023).

C.2 Implementation Details

We use OpenAI API and Huggingface (Wolf et al., 2020), as of January 2024, with specific settings of $temperature$ set to 0 and top_p set to 1 for evaluation of the LoCoMo benchmark. All experiments, including those for RAG-based models, MiniGPT-5 training, and inference, are conducted on an Nvidia A6000 server with FP32. We report results from a single inference run for each model in our experiments. For MiniGPT-5, we used the hyperparameters recommended in the original codebase and trained our models for 10 epochs, which took approximately 30 hours on a single A6000 GPU.

We use the default implementations of BLEU[11],

[11] https://www.nltk.org/_modules/nltk/translate/bleu_score.html

Image sharing: Prompt for generated image caption → image query

```
Let's write short image search queries from textual descriptions of photos shared by a user.
Queries should not include names of people, years and other irrelevant details. For example:

Input: That sounds relaxing, Jeremy! As for video game suggestions, have you ever tried "The Legend
of Zelda: Breath of the Wild"? It's an open-world adventure game that I absolutely love. [shares a
photo of Link standing in front of a breathtaking landscape] Have a look at this stunning view!
Output: the legend of zelda: breath of wild link landscape

Input: {generated_image_caption}
Output:
```

Image response: Prompt for shared image caption → response grounded in image

```
{speaker_1_persona}
{speaker_2_persona}

{speaker_1_name} says, {current_turn}, and shares a photo of {shared_image_caption_blip2}. Write
the most natural question or comment {speaker_2_name} can include in their response.
```

Figure 10: **Prompts for image-sharing and image-response behavior.** The prompt used to convert a caption generated by the virtual agent into an image query for the web-based image crawler in our pipeline (top), and the prompt used to generate a response grounded in the image shared by a virtual agent during a conversation as well as the personas of the respective speakers (bottom).

Category	top-k	BLEU-1/2	Rouge-L	MM-R
Base	-	57.1 / 34.2	12.4	56.1
+ summary	1	58.2 / 34.1	12.8	56.9
+ summary	2	56.5 / 32.8	12.1	55.1
+ summary	5	56.1 / 32.5	12.0	55.2
+ observation	5	**59.7 / 35.1**	**13.6**	**57.8**
+ observation	10	59.1 / 34.9	12.8	57.1
+ observation	25	58.5 / 34.2	12.0	56.5

Table 6: **Multi-modal dialogue generation performance** comparison between different training variants of MiniGPT-5. The optimal performance is shown in **bold**.

ROUGE[12], BertScore[13], FactScore[14] metrics in their respective Python packages in our evaluation protocol.

D Results

D.1 Event Summarization Task

See an example of the five broad categories of event summarization errors made by LLMs, outlined in Section 6.2, in Table 7.

D.2 Multimodal Dialog Generation Task

Results from evaluation of various version of MiniGPT-5 model on the multimodal dialog generation task in the LoCoMo benchmark is in Table 6.

[12] https://pypi.org/project/rouge/
[13] https://pypi.org/project/bert-score/
[14] https://github.com/shmsw25/FActScore